

Jaafar Alhourani

Email : jalhourani3@gmail.com — Phone : +963 0980 329 517 — Address : Homs, Syria

Website: <https://jaafar-a-l-hourani.onrender.com>

ORCID: <https://orcid.org/0009-0001-1541-9942>

GitHub: <https://github.com/AL-Hourani>

Personal Profile

Recent Software Engineering graduate from the University of Homs with strong foundations in computer science, machine learning, and data analysis.

Research Interests

Research interests focus on artificial intelligence, machine learning, reliable AI, and intelligent decision-making systems, with applications in healthcare and real-world data-driven environments.

Education

University of Homs: BSc Computer Science

Sep 2020 – Sep 2025

Major / Specialization: Software Engineering & Information Systems

GPA : 3.1 / 4.0 (Equivalent to 75.96%)

Relevant Modules : Software Engineering , Artificial Intelligence , Database Systems , Data Structures & Algorithms , Machine Learning , Probability and Statistics

Technical Skills

Programming Languages: Python, JavaScript, Go

Data Analysis & Visualization: Skilled in Python data analysis libraries (Pandas, NumPy, Matplotlib) for processing and interpreting datasets.

Machine Learning & AI: Knowledgeable in machine learning using Scikit-Learn; capable of building predictive models and analyzing patterns.

Web & Interactive UI Development: Experienced in building interactive interfaces using React framework.

Database Design & Management: Proficient in designing, implementing, and managing relational databases.

Rapid Learning & Adaptability: Highly self-motivated learner with the ability to quickly master new technologies, frameworks, and tools, and effectively apply them to research and development projects.

Teamwork & Collaboration: Strong ability to work effectively in research and development teams, contributing to problem-solving and project delivery.

Academic Projects

Diabetes Management System (Graduation Project)

Sep – 2025

Developed a comprehensive system to assist diabetes patients in monitoring their health and improving their care. The system includes three versions:

Healthcare Centers : Allows staff to register patients, track health metrics, and log data over time during visits

Supervisor : Monitors medication inventory across centers and responds to inquiries

Patient : Enables patients to track their own health data and receive guidance

Technologies Used : React (UI/UX design), Progressive Web App (PWA) for cross-platform access, Go (API development), PostgreSQL (database management) , Machine Learning

Machine Learning & Medical AI Projects

2025 – Present

Designed and implemented multiple machine learning and deep learning solutions across real-world and research-oriented applications, with a growing focus on medical image analysis and explainable AI systems.

Brain Tumor Detection (CNN-Based) – Developed a deep learning model to classify brain MRI images into tumor and non-tumor categories using Convolutional Neural Networks (CNNs). Applied image preprocessing, feature extraction, and model optimization techniques to improve classification performance.

Medical Anomaly Detection & Patient Risk Monitoring – Built an intelligent system for detecting abnormal patterns in clinical data using unsupervised learning techniques. Implemented real-time risk monitoring and interactive 3D visualizations using Streamlit.

Explainable AI Dashboard – Designed an interactive dashboard to interpret machine learning predictions using SHAP and LIME, enabling transparent model analysis and feature importance visualization.

Explainable AI for Chronic Kidney Disease (CKD) – Developed a full-stack machine learning system for predicting chronic kidney disease using clinical datasets. Integrated explainability tools (SHAP) and deployed results through an interactive Streamlit interface.

Core Machine Learning Projects: - Handwritten Digit Recognition (classification) - House Price Prediction (regression) - Classification projects - Various classification and clustering tasks across multiple datasets

Technologies: Python, Scikit-Learn, TensorFlow/Keras, Pandas, NumPy, Matplotlib, Streamlit, SHAP

Research Experience

Conducted independent, research-oriented work in machine learning and biomedical data science, involving the application of advanced analytical methods to real-world clinical and biological datasets.

Research Publications

Is Systemic Lupus Erythematosus Truly a Single Disease? A Transcriptome-Based Interferon Endotyping Framework

Alhourani, Jaafar Zenodo

<https://doi.org/10.5281/zenodo.19948992>

This study investigates molecular heterogeneity in systemic lupus erythematosus (SLE) using transcriptomic data (GSE10325). Differential gene expression, PCA, UMAP, and functional enrichment analysis reveal strong activation of type I interferon signaling pathways. Patients were stratified into IFN-high and IFN-low molecular endotypes based on an interferon signature score. The results support the hypothesis that SLE is not a single homogeneous disease but a spectrum of interferon-driven molecular subtypes.

Pathway-Based Survival Modeling Reveals Prognostic Signatures in Glioblastoma

Alhourani, Jaafar

<https://doi.org/10.5281/zenodo.19713554>

This study proposes a pathway-based survival modeling framework for glioblastoma using TCGA transcriptomic data. Gene expression data are transformed into pathway activity scores, and Cox proportional hazards models are used to identify prognostic pathways. The resulting model stratifies patients into high-risk and low-risk groups with significant differences in overall survival.

Languages

Arabic: Native

English: Very Good (Academic and Technical)